

Tyrone Marhguy

Computer Engineering — Software Engineering / Systems / Backend / Developer Tools

+1 (267) 320-2734 | tmarhguy@seas.upenn.edu | tmarhguy.com | linkedin.com/in/tmarhguy | github.com/tmarhguy

SUMMARY

Computer Engineering student building systems from search/indexing and transactional storage to compilers, developer tools, and production APIs. Shipped public software and upstream C++/Python fixes to major EDA codebases; strongest in Rust/C++, backend systems, performance, and correctness-focused engineering.

EDUCATION

University of Pennsylvania

B.S.E. Computer Engineering

Philadelphia, PA

Aug. 2024 – May 2028

EXPERIENCE

Software Engineer Intern

May 2026 – Aug. 2026

Aragorn AI

Remote

- Shipped **production backend services and REST APIs** for AI integrations; resolved cross-service failures blocking releases and hardened service boundaries through architecture/code reviews.

Hardware & Firmware Engineer

May 2026 – Aug. 2026

Vero Electric

Philadelphia, PA

- Developed **embedded C/C++ fault-detection firmware** for a BQ79616 battery-monitoring system and debugged firmware/hardware integration on a **240 kWh / 120 kW** platform with lab instrumentation.

SELECTED SOFTWARE & SYSTEMS PROJECTS

SeaLion — Full-Text Search Engine | GitHub

Rust / BM25 / Search / CLI

- Built a full-text search engine from first principles with **Unicode analysis, BM25-ranked phrase search, persistent .seal segments, index generations, and segment merge**; validated behavior through an oracle-backed CLI and **89 passing tests**.

FigDB — Transactional Database | GitHub

Rust / LSM / WAL / Tokio

- Implemented ordered KV storage, a **checksummed WAL**, WAL-backed memtable, SSTables, LSM flush/merge, manifest management, and compaction from first principles; **59 tests passing**.

Lobster — Systems Language & Optimizing Compiler | GitHub

Rust / Compilers / Tomato32

- Designed a statically typed systems language/compiler for dual-LUT3 and Tomato targets; implemented a lexer, **Pratt parser**, name resolution, and type checking behind **lobster check** across **nine Rust crates**.

FramePort — HDMI Capture Inside VS Code | Open VSX | GitHub

TypeScript / VS Code API / FFmpeg

- Published a VS Code extension for live HDMI/USB capture, pixel inspection, PNG screenshots, and H.264 recording; reached **1,250+ Open VSX downloads** and built backpressure, lifecycle recovery, save safety, diagnostics, and encode/decode regression coverage.

Envelop — Browser-to-FPGA Remote Compute | Website | GitHub

TypeScript / Swift / BLE

- Built a remote-compute link from the browser to FPGA hardware: verified native BLE bridges reach physical Tomato boards with explicitly labeled virtual execution as fallback; durable queue/job backend, bridge leases, replay-safe handling, and binary ENVELOP/1 protocol with web, Swift, and Android clients.

Orange Metrics API — Hardware PPA Ingestion Service | GitHub

Python / FastAPI / PostgreSQL / Docker

- Built a Dockerized FastAPI/PostgreSQL service and CLI that discovers and parses Vivado/OpenLane PPA reports, upserts normalized runs, exposes filterable REST endpoints, and ships with Alembic migrations and pytest fixtures.

OPEN-SOURCE ENGINEERING

OpenROAD, Verilator, LibreLane, OpenFPGA | Merged upstream

C++ / Python / EDA Infrastructure

- Merged upstream fixes to production repositories: repaired an **OpenROAD LEF58/ODB parser**, corrected **Verilator Linux peak-memory accounting**, and restored LibreLane/Yosys compatibility and reporting; LibreLane fixes shipped in releases.

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, TypeScript/JavaScript, Swift, Java, Bash, SQL, Embedded C

Systems / Backend: Linux, concurrency, networking, storage engines, compilers, search/indexing, FastAPI, Node.js, REST APIs, PostgreSQL, Supabase, Redis, binary protocols

Tooling / DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, Nginx, FFmpeg, VS Code Extension API, pytest, browser/integration testing